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Contact-Grounded World Models for Robot Manipulation

ABSTRACT

We introduce multi-task Visuo-Tactile World Models (VT-WM), which capture the physics of contact through touch reasoning. By complementing vision with tactile images, VT-WM better understands robot-object interactions in contact-rich tasks, avoiding common failure modes of vision-only models under occlusion or ambiguous contact states, such as objects disappearing, teleporting, or moving in ways that violate basic physics. Trained across a set of contact-rich manipulation tasks, VT-WM improves physical fidelity in imagination, achieving 33% better performance at maintaining object permanence and 29% better compliance with the laws of motion in autoregressive rollouts. Moreover, experiments show that grounding in contact dynamics also translates to planning. In zero-shot real-robot experiments, VT-WM achieves up to 35% higher success rates, with the largest gains in multi-step, contact-rich tasks. Finally, VT-WM shows data efficiency when targeting a new task, outperforming a behavioral cloning policy by over $3.5\times$ in success rate with limited demonstrations.

![Figure 1: Visuo-Tactile World Model complements vision with touch, providing contact grounding of robot-object interactions. The diagram shows two parallel paths for world model planning. The top path, labeled 'Vision-only World Model (V-WM)', shows a sequence of images where a robot arm is stacking cubes. A red dashed box highlights a step where the robot is holding a blue cube, and a red arrow points to it with the text 'Check blue cube'. The bottom path, labeled 'Visuo-Tactile World Model (VT-WM)', shows the same sequence of images, but the robot is holding a different cube (a yellow one) in the same step, indicating a hallucination. The diagram illustrates that the VT-WM has a notion of object permanence for the blue cube when transporting, placing, and releasing it, while the V-WM does not.](8642df2e3828b25d27362bec6d5a0eae_img.jpg)

Figure 1: Visuo-Tactile World Model complements vision with touch, providing contact grounding of robot-object interactions. The diagram shows two parallel paths for world model planning. The top path, labeled 'Vision-only World Model (V-WM)', shows a sequence of images where a robot arm is stacking cubes. A red dashed box highlights a step where the robot is holding a blue cube, and a red arrow points to it with the text 'Check blue cube'. The bottom path, labeled 'Visuo-Tactile World Model (VT-WM)', shows the same sequence of images, but the robot is holding a different cube (a yellow one) in the same step, indicating a hallucination. The diagram illustrates that the VT-WM has a notion of object permanence for the blue cube when transporting, placing, and releasing it, while the V-WM does not.

Figure 1: Visuo-Tactile World Model complements vision with touch, providing contact grounding of robot-object interactions. Notice that when using the WMs for planning a cube stacking task, the VT-WM has notion of object permanence of the **blue cube** when transporting, placing and releasing the object. The contact grounding provided by the vision-based tactile sensor helps to reduce hallucinations often present in V-WMs, enabling more reliable zero-shot planning in contact-rich manipulation tasks.

1 INTRODUCTION

World models (WMs) have emerged as a leading paradigm in machine learning, offering robots the ability to understand the physical world and plan interactions in *imagination* (Agarwal et al., 2025; Russell et al., 2025; Liao et al., 2025). In this work, we advance world models for robot manipulation by extending beyond purely visual imagination to incorporate modalities that directly ground contact interactions (fig. 1). By complementing vision with touch, world models for robot manipulation gain access to local contact signals that anchor predictions in the physics of contact. Tactile sensing provides this crucial information, enabling the model to capture object permanence and force-driven motion, and to move beyond the ambiguities and aliasing of vision alone.

We introduce the first multi-task Visuo-Tactile World Model (VT-WM). Vision provides global context about the robot's kinematics and the task scene, but it does not reveal the state of

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physical contact. Tactile sensing supplies this missing local signal, capturing how the hand and object actually interact. Together, these modalities enable the model to maintain object permanence even under heavy occlusion or visual aliasing. As shown in fig. 1, VT-WM consistently represents the cube throughout the phases of a stacking task: in-hand during transport and placement, and back in the scene once released. Multimodality also disambiguates visually similar states with different outcomes. For example, from the camera view the robot hand may appear to rest on a cloth, yet only tactile feedback can reveal whether contact is sufficient for the cloth to move when wiping, or if it will remain in place. By grounding imagination in both global vision and local touch, VT-WM preserves object permanence and predicts object-robot interactions that respect the laws of motion.

We evaluate the gains of VT-WM over V-WM on a set of contact-rich manipulation tasks. Our first focus is how visuo-tactile training improves imagination by preserving object permanence and adhering to physical motion laws during autoregressive rollouts. Tactile grounding helps prevent common hallucinations in vision-only models, such as objects disappearing under occlusion, teleporting, or moving without applied forces due to visual aliasing. We then assess how this grounding translates to planning. While V-WM and VT-WM perform similarly on reaching tasks that mainly test kinematic fidelity, zero-shot plans generated by VT-WM show a stronger ability to maintain contact in the real world. This capability proves crucial for manipulation actions such as pushing, wiping, and placing, where reliable hand-object interaction determines task success.

Our contributions are threefold:

- We propose the first multi-task visuo-tactile world model that integrates fingertip tactile sensing with vision to jointly model global context and local contact dynamics.
- We show that visuo-tactile grounding substantially improves imagination quality, achieving a 33% gain in object permanence and a 29% gain in compliance with physical laws, evaluated across a set of manipulation tasks.
- We demonstrate that these improvements in imagination enable more reliable zero-shot planning on real robots, with up to 35% higher success in contact-rich tasks.

2 RELATED WORKS

In this work we aim to train general (multi-task) world models that can leverage both tactile and visual observations. Specifically we train latent-state world models, which first project observations into latent representations and then train an action-conditioned dynamics model in that latent space.

Foundational encoders for vision and touch: Unlike the established hardware platforms for computer vision, the field of robotic tactile sensing lacks a single standardized sensor. Nevertheless, vision-based tactile sensors have emerged as one of the most prominent solutions. Devices like GelSight (Yuan et al., 2017), Digit (Lambeta et al., 2020), and the more recent Digit 360 (Lambeta et al., 2024) capture tactile information by imaging the deformation of a soft elastomer surface.

Similar to the development of general-purpose visual encoders like CLIP (Radford et al., 2021), DINO (Caron et al., 2021; Oquab et al., 2023), I-JEPA (Assran et al., 2023), and Cosmos Tokenizer (Agarwal et al., 2025), recent efforts have focused on creating foundational models for vision-based tactile sensors through self-supervised learning. Models such as SITR (Gupta et al., 2025), T3 (Zhao et al., 2024), UniT (Xu et al., 2025), Sparsh (Higuera et al., 2024), and Sparsh-X (Higuera et al., 2025) learn robust, low-dimensional tactile representations without requiring explicit labels. Benchmarks like TacBench (Higuera et al., 2024) evaluate the quality of these embeddings, demonstrating their ability to compress information about contact dynamics, including force fields, slip states, and pose changes, as well as static properties like texture and material. In this work, we use Digit 360 sensors as fingertips for an Allegro Hand mounted on a Franka Panda arm and we use the Sparsh-X model (Higuera et al., 2025) to obtain tactile embeddings, and the Cosmos encoder (Agarwal et al., 2025) to obtain RGB embeddings.

Action-Conditioned World Models for Real World Robotics: There has been an influx of training general purpose action-conditioned video-generation models (Hu et al., 2023; Russell et al., 2025; Yang et al., 2024; Bruce et al., 2024; Agarwal et al., 2025; Assran et al., 2025). However, these lines of work focus on show-casing the generation capabilities, and only have limited (if any) results for using these models to control robots.

The majority of previous work on world models applied to real world tasks focuses on visual dynamics models (Agrawal et al., 2016; Byravan et al., 2017; Das et al., 2020; Nagabandi et al., 2020;

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Finn et al., 2016; Ebert et al., 2017; 2018; Yen-Chen et al., 2020). Visual dynamics models are either trained directly in pixel-space (Finn et al., 2016; Ebert et al., 2017; 2018; Yen-Chen et al., 2020; Alonso et al., 2024), or in a learned latent space (Watter et al., 2015; Agrawal et al., 2016; Ha & Schmidhuber, 2018; Hafner et al., 2019; Nair et al., 2022; Wu et al., 2023; Tomar et al., 2024; Hu et al., 2024; Lancaster et al., 2024), or in more structured representation spaces such as keypoint representations (Manuelli et al., 2020; Das et al., 2020) or tracked 3D states (Nagabandi et al., 2020). In our work, we train action-conditioned world models in latent states extracted from both RGB and tactile observations.

While several works have investigated learning dynamics model on touch observations (Sutanto et al., 2019; Tian et al., 2019; Ai et al., 2024) there is little work on training world models with vision and touch (Zhang & Demiris, 2023). Furthermore these dynamics models are task-specific - while in our work we aim to train general purpose multi-modal world models that can be used for visual MPC on multiple tasks.

3 WORLD MODELS THAT UNDERSTAND CONTACT

Vision-only world models have shown promising capabilities in action steerability and spatial reasoning (Assran et al., 2025; Agarwal et al., 2025), generating plausible rollouts with reasonable robot kinematics and high-quality visuals. These properties make them useful for planning free-space motions. However, simulating object interactions comes with some challenges. Object motion depends on forces invisible to exocentric cameras, and occlusions during grasping, pushing, or placing often cause artifacts such as teleportation, disappearance, or physically implausible dynamics.

We introduce *Visuo-Tactile World Model* (VT-WM), which uses tactile sensing to complement vision to overcome these limitations. Touch provides contact information during occlusion, grounding the model’s imagination in contact physics and producing more accurate rollouts for contact-rich manipulation tasks.

3.1 WHAT VISION DOESN’T SEE: SENSING CONTACT WITH TOUCH

Touch provides essential local perception, enabling robots to distinguish properties like stiffness, friction, and roughness that are difficult to infer from vision alone. It also captures the dynamics of contact, which is crucial for manipulation tasks. For instance, when manipulating an object in-hand, touch provides context about forces, slip, and subtle pose changes.

Vision-based tactile sensors typically stream image data at 30-60 FPS, providing rich information about the contact area, including force, shape, and texture features. This tactile information is crucial for disambiguating contact states that from an exocentric camera may appear visually similar. For example, a robot hand’s grasp on a cup might look the same from a distance, but tactile sensing can differentiate between a no-contact state, a subtle touch, or a firm grasp. In this work, we use Digit 360 sensors as fingertips for an Allegro Hand mounted on a robot arm. A visualization of tactile images captured by a Digit 360 sensor is shown in fig. 2.

![Figure 2: Tactile images from Digit 360 sensors. The figure shows a grayscale image of a robotic hand holding a white cup. Overlaid on the hand are four circular heatmaps representing tactile data from Digit 360 sensors. Each heatmap shows a different color intensity (blue, green, yellow, red) indicating the level of contact or force. White dashed boxes highlight specific areas of high contact, such as the fingertips and the thumb.]

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Figure 2: Tactile images from Digit 360 sensors. The figure shows a grayscale image of a robotic hand holding a white cup. Overlaid on the hand are four circular heatmaps representing tactile data from Digit 360 sensors. Each heatmap shows a different color intensity (blue, green, yellow, red) indicating the level of contact or force. White dashed boxes highlight specific areas of high contact, such as the fingertips and the thumb.

Figure 2: Tactile images from Digit 360 sensors. White boxes highlight contact while the hand holds a screw.

3.2 MULTITASK VISUO-TACTILE WORLD MODEL

3.2.1 MODEL ARCHITECTURE

Our visuo-tactile world model is designed to address a key challenge in multimodal robot world models: *how to combine exocentric vision with tactile sensing in order to generate consistent imagined futures*. As shown in fig. 3, the architecture consists of three main components: a *vision encoder*, a *tactile encoder*, and an autoregressive *predictor*.

The vision encoder extracts latent states s_k that capture the robot and its environment from exocentric video. The tactile encoder compresses high-frequency contact feedback into a compact state t_k that emphasizes salient physical interactions. These representations are fused with control

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![Diagram of the Visuo-Tactile World Model architecture. The model takes visual and tactile inputs, encodes them using Cosmos and Sparsh-X tokenizers, and processes them through a transformer predictor to generate next-step states. The visual input is a sequence of frames from a camera (15 fps) showing a robot hand interacting with a cup. The tactile input is a sequence of heatmaps from a Digit 360 sensor (6 Hz) showing contact information. The Cosmos Tokenizer encodes visual frames into latent sequences s_{k-7} to s_k. The Sparsh-X Tokenizer encodes tactile heatmaps into latent sequences t_k^{(1000,0)} to t_k^{(1000,3)}. These latents are concatenated and passed through a Predictor (Transition Model) which also takes control actions a_k. The Predictor outputs next-step states s_{k+1} and t_{k+1} via Projector heads.](2fa4a1bf91d0f34e87c689fbc1211fe3_img.jpg)

Diagram of the Visuo-Tactile World Model architecture. The model takes visual and tactile inputs, encodes them using Cosmos and Sparsh-X tokenizers, and processes them through a transformer predictor to generate next-step states. The visual input is a sequence of frames from a camera (15 fps) showing a robot hand interacting with a cup. The tactile input is a sequence of heatmaps from a Digit 360 sensor (6 Hz) showing contact information. The Cosmos Tokenizer encodes visual frames into latent sequences s_{k-7} to s_k. The Sparsh-X Tokenizer encodes tactile heatmaps into latent sequences t_k^{(1000,0)} to t_k^{(1000,3)}. These latents are concatenated and passed through a Predictor (Transition Model) which also takes control actions a_k. The Predictor outputs next-step states s_{k+1} and t_{k+1} via Projector heads.

Figure 3: Visuo-Tactile World Model. Vision (s_k) and tactile (t_k) latents, obtained from Cosmos and Sparsh-X encoders, are processed by a transformer predictor given control actions a_k to generate next-step states (s_{k+1}, t_{k+1}) .

actions and passed to the predictor, a forward dynamics model that estimates the next-step states $(s_{k+1}, t_{k+1}) \sim \mathcal{P}_\phi(s_k, t_k | a_k)$.

This formulation enables the model to *imagine multiple possible futures under control actions*. Unlike purely visual world models, our predictor leverages tactile signals to disambiguate perceptually identical visual states. For instance, two identical video frames of a robot hand around a cup can lead to different rollouts: if the tactile input indicates contact, the imagined sequence shows the cup being lifted; if not, the cup remains on the table. This tactile-informed disambiguation is significant for planning in contact-rich manipulation.

We frame the predictor as a supervised next-state estimation problem with ground-truth future latents as targets. Both modalities are encoded with pretrained networks: Cosmos tokenizer (Agarwal et al., 2025) for vision and Sparsh-X (Higuera et al., 2025) for Digit 360 tactile sensors. The encoded contexts s_k and t_k are augmented with sinusoidal positional embeddings, and projected into a unified representation $\mathbb{R}^{(b,t,s,d)}$. Vision and tactile tokens are concatenated along the spatial dimension to form a unified input sequence. The predictor then processes these multi-modal tokens through a 12-layer transformer that alternates between two types of attention mechanisms:

Spatio-Temporal Self-Attention The model processes vision-touch tokens through factorized attention that operates in two stages: **spatial** attention enables all tokens within a timestep to interact, while **temporal** attention tracks how each token evolves across past timesteps. This factorization efficiently captures both local dynamics and global context while avoiding the $O((THW)^2)$ complexity of full spatiotemporal attention. Action tokens undergo the same factorized attention process.

Action Conditioning via Cross-Attention After each self-attention block, vision-touch tokens cross-attend to action tokens to incorporate the robot's control inputs into the predictions. This alternating pattern of self-attention and cross-attention allows the model to iteratively refine its latent states based on both sensory observations and executed actions.

All attention layers employ Rotary Position Embeddings (RoPE) (Su et al., 2023) for relative position encoding. After the transformer, the representations are projected back to their original dimensions through modality-specific output heads, yielding predictions s_{k+1} and t_{k+1}.

3.2.2 TRAINING VISUO-TACTILE WORLD MODEL

During training, the vision input is a 1.5-second exocentric videoclip (9 frames at 6 fps, 320×192 resolution), encoded framewise with Cosmos. The tactile input consists of two frames per Digit 360

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sensor (four sensors total), covering the most recent 0.16 seconds. This shorter horizon reflects the higher temporal frequency and local nature of contact information, which complements the slower, global context provided by vision. The action input includes changes in proprioceptive state (translation, quaternion rotation) and a binary hand state representing pre-set open/close configurations. We chunk action sequences from 30Hz into groups of 5, with the full chunk of delta-states provided to the predictor. This combination ensures that the predictor models both the external scene and the internal actuation history, a prerequisite for accurate visuo-tactile imagination. The model uses a maximum context length of 9 frames for both vision and touch modalities. Additional details about hyperparameters and training dataset are provided in appendix A.

The model is trained with an objective that combines teacher forcing with autoregressive sampling to balance training stability with long-horizon coherence as proposed by Assran et al. (2025).

Teacher Forcing Loss. The primary training signal comes from next-step prediction with ground-truth context. Given a sequence of T frames, we compute:

$$L_{teacher} = \sum_{k=1}^{T-1} \|\hat{s}_{k+1} - s_{k+1}\|_1 + \|\hat{t}_{k+1} - t_{k+1}\|_1 \quad (1)$$

where $\hat{s}_{k+1}$ and $\hat{t}_{k+1}$ are predicted from ground-truth states up to time k, and s_{k+1} and t_{k+1} are the encoded latents given ground truth observations at time step $k+1$. This provides dense supervision and stable gradients but can lead to distribution shift during autoregressive rollouts.

Sampling Loss. To improve long-horizon generation, we additionally train on sampled trajectories. During training, we sample future states autoregressively for H steps (typically $H = 3 - 5$), then compute predictions conditioned on these sampled states:

$$L_{sampling} = \sum_{k=1}^H \|\hat{s}_{k+1}^{sampled} - s_{k+1}\|_1 + \|\hat{t}_{k+1}^{sampled} - t_{k+1}\|_1 \quad (2)$$

The sampled states are generated without gradients to prevent training instability. The final loss combines both objectives with equal weighting: $L = L_{teacher} + L_{sampling}$.

3.2.3 PLANNING IN IMAGINATION

The action-conditioned nature of our predictor enables the use of the visuo-tactile world model as a simulator within a Cross-Entropy Method (Rubinstein, 1997) (CEM). At each step, the planner samples a population of action sequences $\{a_i^{k:k+H}\}_{i=1}^N$ over a horizon H. For each sequence, the predictor autoregressively generates future latents $(\hat{s}_{k+1:k+H}, \hat{t}_{k+1:k+H})$. A cost function, defined by energy minimization with respect to a goal image, assigns a score to each trajectory. In practice, this cost can be as simple as an ℓ_2 distance between the final predicted visual latent $\hat{s}_{k+H}$ and the latent of the goal image s_{goal}. CEM then selects the top-performing fraction of sequences, updates the sampling distribution toward them, and iterates until convergence. The best sequence is then executed on the real robot in an open-loop manner.

We do not provide the tactile modality as a goal signal, thus the planning objective remains purely vision-based. The role of tactile in the VT-WM is to enhance the reliability of the learned world model and, therefore, to improve planning indirectly. First, tactile feedback during training enables the world model to capture contact physics that are difficult to infer from vision alone. Second, when generating rollouts, tactile context in the initial state helps disambiguate visually identical observations (e.g., distinguishing whether the robot is already in contact with an object). This yields more physically consistent imagined futures and more accurate cost evaluations, which we hypothesize translate into higher-quality plans.

4 EXPERIMENTS

Our experimental framework evaluates the advantages of visuo-tactile world model (VT-WM) for robot manipulation by addressing the key questions:

- **Contact Perception:** Do VT-WMs better capture object permanence and causal compliance than vision-only WMs, and generate futures consistent with action conditioning?

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- *Zero-shot Planning*: Does improved contact perception lead to more reliable zero-shot plan transfer in open-loop execution?
- *Data Efficiency*: Given limited demonstrations, how does fine-tuning a multi-task world model for planning compare to behavioral cloning?

4.1 CONTACT PERCEPTION

To evaluate the benefits of incorporating touch, we compare rollouts from a multi-task vision-only world model (V-WM) and our multi-task visuo-tactile world model (VT-WM), conditioned on the same actions and context. The action sequences are drawn from successful demonstrations on the real system, which enables a direct comparison between each model's rollouts and the corresponding ground-truth videos. This setup allows us to assess how well the models capture motion dynamics and the physical plausibility of object interactions. We focus our evaluation on object permanence, causal compliance, and action controllability. These metrics are components of the World Consistency Score (Rakheja et al., 2025), a proposed benchmark in the state-of-the-art to evaluate a generative model's ability to maintain coherent and physically plausible futures over time.

Object Permanence: This metric assesses a model’s ability to maintain a consistent representation of an object’s existence and state even when the object is temporarily occluded. As shown in fig. 5, we evaluate whether objects remain represented during heavy occlusion (e.g., during a grasp) and reappear in the correct state once revealed. In the cube-stacking task, VT-WM exhibits stronger object permanence than V-WM: as the blue cube is occluded in the hand during transport and placement, VT-WM preserves its representation, and upon release, the cube re-emerges in the imagined scene at the correct location above the yellow target cube. In appendix B-fig. 13, we visualize the tactile predictions generated by VT-WM, demonstrating the model’s ability to maintain congruent visual and tactile representations of object contact.

For quantitative evaluation, we employ CoTracker (Karaev et al., 2024) to track keypoints on the object, providing pixel-level visibility and trajectories. We then compare the normalized Fréchet distance between the ground-truth visual trajectory and the one imagined by V-WM and VT-WM under the ground-truth action conditioning. To ensure comparability, trajectories are expressed relative to the object’s initial image position and normalized by the length of the ground-truth trajectory. A lower Fréchet distance indicates that the imagined trajectory more closely reflects the real motion and state of the object, thereby capturing the physical coherence required for object permanence.

Fig. 4 reports normalized Fréchet distances across five tasks. To assess statistical significance, we complement the Fréchet distance analysis with paired *t*-tests across tasks. VT-WM achieves consistently lower distances than V-WM, with statistically significant improvements in *place fruits* ($t = 4.38, p < 0.001$), *push fruits* ($t = 6.06, p < 10^{-6}$), and *cube stacking* ($t = 2.40, p < 0.05$). For *wipe with cloth* and *scribble with marker*, the differences follow the same downward trend but do not reach significance at the 5% level. Quantitatively, VT-WM reduces normalized Fréchet distance by 18–47% across all five tasks, corresponding to an average overall reduction of $\approx 33\%$. These results indicate that VT-WM produces more physically coherent rollouts, with statistically significant gains in tasks requiring reliable object permanence, such as object pushing and stacking.

Causal Compliance: This metric evaluates whether changes in object state occur as physically plausible consequences of the robot’s actions. A causally compliant model predicts that an object’s state changes only when it is subject to external forces. Assessing causal compliance is essential for developing physics-informed world models that respect the principles of contact dynamics and avoid unrealistic motions or deformations.

For a quantitative measure of causal compliance, we use CoTracker to compute the trajectory error of keypoints on objects in the scene that are not subject to any external force and should therefore

![Bar chart showing Normalized Fréchet Distance for five tasks: Place fruits, Push fruits, Wipe with cloth, Stack cubes, and Scribble with marker. Two models are compared: V-WM (teal) and VT-WM (purple). VT-WM consistently shows lower distances than V-WM across all tasks, with the most significant improvements in Push fruits and Stack cubes.](92c6488dfdf125bc1b7bb6a235394652_img.jpg)

Task	V-WM (Normalized Fréchet Distance)	VT-WM (Normalized Fréchet Distance)
Place fruits	~0.05	~0.03
Push fruits	~0.50	~0.25
Wipe with cloth	~0.40	~0.30
Stack cubes	~0.50	~0.28
Scribble with marker	~0.15	~0.10

Bar chart showing Normalized Fréchet Distance for five tasks: Place fruits, Push fruits, Wipe with cloth, Stack cubes, and Scribble with marker. Two models are compared: V-WM (teal) and VT-WM (purple). VT-WM consistently shows lower distances than V-WM across all tasks, with the most significant improvements in Push fruits and Stack cubes.

Figure 4: *Object permanence*. VT-WM achieves an average reduction of $\approx 33\%$ relative to V-WM (with 95% CI) of the normalized Fréchet distances for objects in motion.

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![Figure 5: A grid of images showing robot arm actions across five tasks: Transport cube, Place cube, Release cube, Reach fruit, and Push fruit. The grid has three rows: Ground Truth, VT-WM, and V-WM. The Ground Truth row shows successful actions. The VT-WM row shows successful actions with some minor object deformations. The V-WM row shows various failures: 'Object less visible', 'No object permanence', 'No object permanence', 'Slight object deformation', and 'Severe object deformation'.](9ebd85380ef496499e5f23ec0d9cd744_img.jpg)

Figure 5: A grid of images showing robot arm actions across five tasks: Transport cube, Place cube, Release cube, Reach fruit, and Push fruit. The grid has three rows: Ground Truth, VT-WM, and V-WM. The Ground Truth row shows successful actions. The VT-WM row shows successful actions with some minor object deformations. The V-WM row shows various failures: 'Object less visible', 'No object permanence', 'No object permanence', 'Slight object deformation', and 'Severe object deformation'.

Figure 5: VT-WM preserves object permanence and consistent hand–object interactions during imagination, while V-WM often loses objects or produces severe deformations.

remain stationary. We again use the normalized Fréchet distance between ground-truth and imagined rollouts as our metric. A higher Fréchet distance indicates that the world model hallucinates changes in the position or deformation of these passive objects, thereby violating basic physical laws such as Newton's first law of motion. As shown in fig. 6, VT-WM consistently achieves lower distances than V-WM across most tasks. Paired t -tests confirm statistically significant improvements at the 5% level in *place fruits* ($t = 3.66, p < 0.001$), *push fruits* ($t = 2.28, p < 0.05$), and *wipe with cloth* ($t = 2.99, p < 0.01$), while differences in *cube stacking* ($t = 1.75, p = 0.09$), and *scribble with marker* ($t = -1.22, p = 0.23$) are not significant. Quantitatively, VT-WM achieves relative reductions of 43.6%, 16.4%, and 66.1% in *place fruits*, *push fruits*, and *wipe with cloth*, respectively, alongside a smaller improvement in *cube stacking* and a degradation in *scribble with marker*. Overall, VT-WM reduces hallucinated motion by an average of $\approx 29\%$ across tasks, reflecting stronger causal compliance in most scenarios.

In fig. 7 we show snapshots of a trajectory where the robot performs a wiping motion just above a cloth, without making contact. In the ground-truth sequence (top row), keypoints on the cloth remain stationary. In contrast, the V-WM's rollout (bottom row), conditioned on the real actions, shows significant displacement of keypoints and deformations of the cloth. This highlights the V-WM's difficulty to distinguish between contact and non-contact states based on visual input alone. The VT-WM's rollouts (middle row), however, exhibit fewer artifacts and less variation, demonstrating the advantage of tactile sensing in providing the world model with critical physical grounding.

In appendix B we showcase the action controllability of the world model. We compare ground-truth trajectories with VT-WM rollouts under the same action sequences and illustrate what the world model *imagines* in terms of contact.

4.2 ZERO-SHOT PLANNING TRANSFER TO REAL-WORLD

Does the improved contact perception of VT-WM translate into superior planning performance on a real robot? We hypothesize that VT-WM produces more effective plans for contact-rich tasks. For example, in a cube-stacking task, a physically grounded model should avoid opening its hand while transporting a cube. Similarly, for pushing, the model must recognize that contact is a prerequisite for object motion.

![Figure 6: A bar chart showing Normalized Fréchet Distance for five tasks: Place fruits, Push fruits, Wipe with cloth, Stack cubes, and Scribble with marker. The chart compares V-WM (teal bars) and VT-WM (purple bars). VT-WM consistently shows lower distances than V-WM across all tasks, indicating better causal compliance.](023b142f90e1253702ac88b18380d3ec_img.jpg)

Task	V-WM (Normalized Fréchet Distance)	VT-WM (Normalized Fréchet Distance)

Place fruits	~0.55	~0.30
Push fruits	~0.35	~0.30
Wipe with cloth	~0.80	~0.25
Stack cubes	~0.35	~0.25
Scribble with marker	~0.35	~0.50

Figure 6: A bar chart showing Normalized Fréchet Distance for five tasks: Place fruits, Push fruits, Wipe with cloth, Stack cubes, and Scribble with marker. The chart compares V-WM (teal bars) and VT-WM (purple bars). VT-WM consistently shows lower distances than V-WM across all tasks, indicating better causal compliance.

Figure 6: *Causal compliance* evaluates WMs adherence to the laws of motion. Normalized Fréchet distance for static objects (95% CI) shows that VT-WM outperforms V-WM, with an overall improvement of $\approx 29\%$.

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![Figure 7: Comparison of rollouts for Ground Truth, VT-WM, and V-WM. The figure is divided into two columns: 'Hand moving above cloth without contact' and 'Hand fails object grasp'. Each column shows a sequence of images for each method. Ground Truth shows no contact or no object change. VT-WM shows object motion or slight deformation. V-WM shows object motion or object disappears.](892f25e3d71d8e315a2a51092a8a8da7_img.jpg)

Figure 7: Comparison of rollouts for Ground Truth, VT-WM, and V-WM. The figure is divided into two columns: 'Hand moving above cloth without contact' and 'Hand fails object grasp'. Each column shows a sequence of images for each method. Ground Truth shows no contact or no object change. VT-WM shows object motion or slight deformation. V-WM shows object motion or object disappears.

Figure 7: Comparison of rollouts, illustrating that VT-WM prevents spurious motion of objects not subject to forces, whereas V-WM often hallucinates unintended displacements.

![Figure 8: Success rate of plans via CEM with VT-WM and V-WM on real robot. The figure includes a bar chart for five tasks and a smaller chart for a new task. The bar chart shows success rates for VT-WM (blue) and V-WM (purple) for Reach Button, Push Fruits, Reach&Push, Wipe Cloth, and Stack Cubes. The new task chart compares BC Policy (22%) and VT-WM (78% and 1250.0%). Below the charts are images of the robot performing the tasks.](d864789b0d8384da1d22fd6a5d76bbdf_img.jpg)

Task	V-WM (%)	VT-WM (%)
Reach Button	100%	100%
Push Fruits	83%	92%
Reach&Push	69%	93%
Wipe Cloth	70%	92%
Stack Cubes	75%	83%

Policy	Success Rate (%)
BC Policy	22%
VT-WM	78% (1250.0%)

Figure 8: Success rate of plans via CEM with VT-WM and V-WM on real robot. The figure includes a bar chart for five tasks and a smaller chart for a new task. The bar chart shows success rates for VT-WM (blue) and V-WM (purple) for Reach Button, Push Fruits, Reach&Push, Wipe Cloth, and Stack Cubes. The new task chart compares BC Policy (22%) and VT-WM (78% and 1250.0%). Below the charts are images of the robot performing the tasks.

Figure 8: *Left*: Success rate of plans via CEM with VT-WM and V-WM on real robot. For all tasks the VT-WM achieves equal or better performance (blue labels), empirically demonstrating the better planning capability with contact grounding via tactile sensing. *Right*: success rate on new task highlights the data efficiency of VT-WM compared to classical behavioral cloning (BC) policies.

To evaluate this, we employ the Cross-Entropy Method (CEM) (Rubinstein, 1997; De Boer et al., 2005) to solve a goal-conditioned energy minimization problem. The objective is to plan an optimal action sequence over a fixed horizon H, where the cost is the distance in latent space between the final predicted visual state s_{k+H} and the goal image latent s^{goal}. The search space for CEM is $\mathbb{R}^7$, consisting of 3D translation and 3D orientation of the wrist pose, plus a binary variable for the hand's open/closed configuration.

We evaluate open-loop zero-shot transfer of the generated plans on the real robot. To initialize planning consistently, the initial RGB and tactile embeddings are passed as context to the world model, indicating whether optimization should begin from an in-contact or no-contact state. Both VT-WM and V-WM are tested on five tasks of increasing difficulty: *reach button*, *push fruits*, *reach & push*, *wipe cloth*, and *stack cubes*. The first two are single-goal tasks, while the latter three involve multiple subgoals.

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Fig. 8(left) reports success rates, averaged over five trials per task from distinct initial conditions. The results confirm the superior planning capability of VT-WM across all tasks, supporting our hypothesis that a contact-aware model generates more effective plans. On simple tasks such as *reach button*, both models achieve 100% success, consistent with prior visual world models for robot manipulation such as V-JEPA-2AC (Assran et al., 2025). However, the benefits of tactile input become increasingly evident in contact-rich tasks: VT-WM improves success rates by 10% on *push fruits*, 35% on *reach & push*, 31% on *wipe cloth*, and 11% on *stack cubes*. These gains are most pronounced in multi-step tasks involving sustained contact, where vision alone is insufficient to inform about the object state during planning. In appendix C, we describe the experimental setup for each task and present a qualitative evaluation of the planned trajectories and their corresponding real-world executions.

4.3 DATA EFFICIENCY

For a new task with a limited number of successful demonstrations, *how does fine-tuning a multi-task world model for planning compare to training a task-specific behavioral cloning (BC) policy?* We hypothesize that VT-WM can extract task structure even in low-data regimes, since it already encodes contact dynamics from prior tasks. For example, insertion may involve a new object, but concepts such as alignment and adjusting contact with a receptacle can be reused. In contrast, a BC policy must learn both spatial and contact reasoning from scratch.

We collect 20 demonstrations of the task "plate in the dish rack," which requires transporting the plate and inserting it between the racks. We augment our multi-task dataset (see appendix A.0.1) with the new sequences and continue training VT-WM, while also training a task-specific BC policy ACT (Zhao et al., 2023) that outputs action chunks over a fixed horizon. For VT-WM, we use CEM planning and zero-shot transfer to the real robot. The task is divided into two subgoals: alignment and insertion (see fig. 8). The BC policy is deployed in closed loop, where at each timestep it receives the latest RGB and tactile inputs and executes the first action of the predicted chunk.

We evaluate both methods in nine real-robot trials, randomizing the initial plate-in-grasp pose. As shown in fig. 8(right), VT-WM planning achieves a 77% success rate, compared to 22% with BC. These results highlight the data efficiency of multi-task world models and their advantage over task-specific policies. Moreover, failure modes differ, for instance, VT-WM mostly places the plate beside the rack, whereas 57% of BC failures involve the robot never reaching the rack at all.

5 DISCUSSION AND CONCLUSION

Visuo-Tactile World Model (VT-WM) leverages tactile sensing to complement vision, enabling world models for robot manipulation to be grounded in contact. While vision-only world models (V-WMs) have shown promise in spatial reasoning and capturing robot kinematics, they are prone to hallucinate object interactions, with failures such as object disappearance, teleportation, or unrealistic deformations. By integrating fingertip tactile sensing with exocentric vision, VT-WM grounds imagination in the physics of contact, producing more accurate rollouts and capturing core concepts such as object permanence under occlusion and compliance with physical laws.

We studied the gains of multimodality in VT-WM over V-WM through three key questions. First, *does adding touch improve a world model's understanding of contact?* Comparing autoregressive rollouts under identical action sequences, we found that VT-WM preserved object permanence, even when objects were occluded by the robot hand, and maintained resting states for objects not subject to external forces. Quantitatively, VT-WM reduced normalized Fréchet distance by 33% on average relative to V-WM, better reflecting true object dynamics.

Second, *does contact grounding improve planning?* Using CEM-based imagination for goal-conditioned planning, we zero-shot transferred plans to a real robot under randomized initial conditions. While both models performed similarly on free-space tasks such as reaching, VT-WM achieved up to 35% higher success rates in contact-rich tasks requiring precise hand-object interaction, including pushing, wiping, and stacking.

Third, *is VT-WM data efficient when targeting a new task?* To compare with task-specific behavioral cloning (BC), we fine-tuned VT-WM on just 20 demonstrations of a plate-insertion task. VT-WM reached a 77% success rate, over $3\times$ higher than BC, by reusing priors from previously learned contact-rich tasks such as alignment and insertion. This highlights its ability to efficiently adapt to new tasks with limited data.

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